

AI-Powered Observability

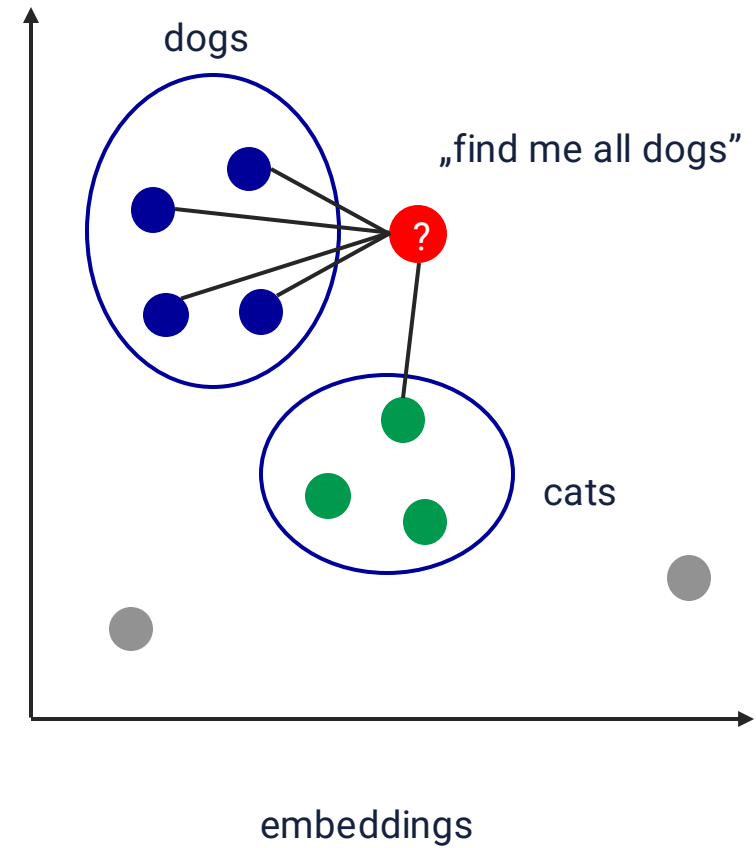
Intelligent Log Monitoring & Root Cause Insight

Ismail Paktiawal

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AGENDA

- 01** Problem & Need
- 02** Stakeholder & Value
- 03** Solution Concept
- 04** Hackathon Delivery



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Chapter 1

Problem & Need

Problem & Need: Logs are abundant, insight is scarce

THE PROBLEM

An LLM cannot use evidence it never sees.



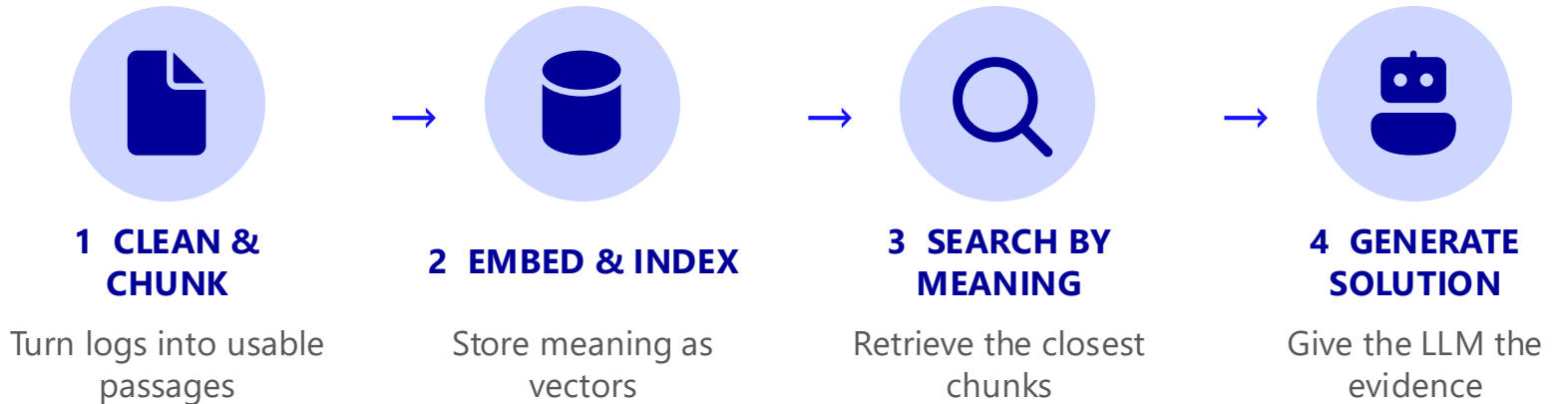
PRIVATE • FAST-CHANGING • UNSTRUCTURED

Raw logs sit outside the model, go stale quickly, and are too large to paste into every prompt.

Result: **generic or unreliable answers**

THE SOLUTION

Retrieve the right evidence before generation.



Question → relevant log chunks → grounded answer

The response is based on retrieved evidence—not on guesswork.

Hackathon need

Can we turn raw logs into a ranked, explainable incident narrative with probable root cause and recommended next action?

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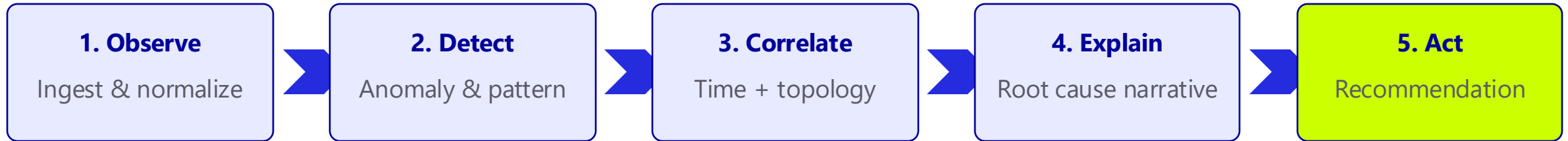
Chapter 2

Goal & Benefit

Target experience and measurable operational outcomes

Goal & Benefit

A single investigation flow from detection to evidence, explanation and decision.



Goal

Reduce cognitive load and shorten the path from “something is wrong” to “this is the likely cause, evidence and next action.”

Expected value & KPIs

Hackathon success should be evaluated by decision quality, not by dashboard count.

Faster triage

Time to first probable cause

Target: compare AI vs. manual

Higher precision

True anomalies in top findings

Measure precision@k

Better explainability

Evidence links per conclusion

Human review score

Lower noise

Duplicate alerts consolidated

Reduction ratio

Reusable assets

Parsers, prompts, evaluation set

Portable prototype

Governed usage

PII handling and access boundaries

Compliance checklist

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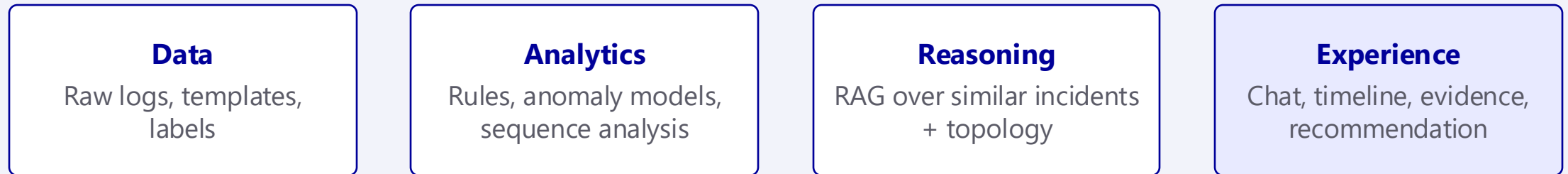
Chapter 3

Solution Concept

Evidence-first architecture, data and analytics approach

Proposed architecture: evidence-first, model-assisted

The LLM explains correlated evidence; deterministic steps retain control over parsing, retrieval and scoring.



Hackathon data foundation: Loghub/Synthetic Data

Use labeled and unlabeled public system logs to test parsing, anomaly detection and explanation.

Dataset	System	Label	Scale
HDFS_v1	Distributed file system	Yes	11.2M lines
BGL	Supercomputer	Yes	4.7M lines
OpenStack	Infrastructure	Yes	207.8k lines
Linux	Operating system	No	25.6k lines

Recommended start

Small to Big

Important: Loghub warns that logs are not always sanitized or anonymized. Treat the prototype as research data and isolate it accordingly.

AI analytics stack: combine methods, do not bet on one model

A layered design improves robustness and keeps explanations traceable.

Log parsing

Convert variable messages into stable templates

Anomaly detection

Score unusual events, sequences and rates

Correlation

Join events by time, entity and service dependency

Retrieval

Find similar historical patterns and known fixes

LLM reasoning

Summarize evidence and generate ranked hypotheses

Guardrails

Redaction, grounding, confidence and abstention

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Chapter 4

Hackathon Delivery

Scope, workstreams and definition of done

Hackathon scope: a thin vertical slice that can be demonstrated

Deliver one credible incident journey, not a broad platform mock-up.

Must have

- Ingest one Loghub dataset
- Parse templates
- Detect one anomaly class
- Evidence timeline
- Ranked root-cause hypothesis

Should have

- Natural-language investigation
- Similar incident retrieval
- Confidence + abstention
- Simple service topology

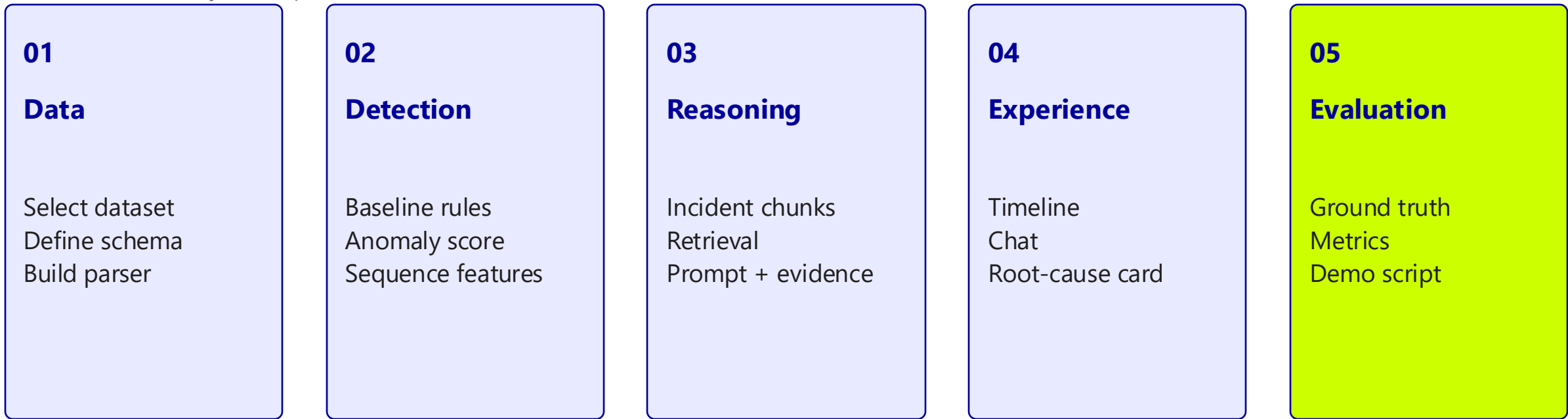
Out of Scope?

- Production auto-remediation
- Enterprise-wide ingestion
- Full vendor replacement
- Universal log parser

Demo story: “A service fails → the system detects the anomaly → correlates the sequence → identifies the likely cause → shows evidence → recommends the next steps/resolution.”

Execution plan and workstreams

Parallelize data, analytics, experience and evaluation from the first hour.



Definition of Done Reproducible demo, evidence-linked output, baseline comparison, documented limitations and clear next-step recommendation.

Key Takeaways

01 Start with a real operational need

Reduce the path from symptom to evidence-backed cause.

02 Build an evidence-first AI pipeline

Parsing, correlation and retrieval before generation.

03 Keep the hackathon narrow

One dataset, one incident class, one convincing demo.

04 Measure decision quality

Speed, precision, explainability and noise reduction.

Decision: proceed with a basic set as the initial benchmark and validate the end-to-end incident journey.



MHP Management- und IT-Beratung GmbH

Film- und Medienzentrum | Königsallee 49 | D-71638 Ludwigsburg

Telefon +49 (0)7141 7856-0 | Fax +49 (0)7141 7856-199

E-Mail info@mhp.com | Internet www.mhp.com